Business Context

Procurement analytics case study | Read this before opening the data

The setting

The company sources raw materials, packaging, spare parts and logistics services through a mix of two buying modes. Most purchase orders go through the regular scheduled process (**BuyingMode = "Contract"**), negotiated in advance under standing terms. The rest are placed opportunistically (**BuyingMode = "Spot"**) when a buyer sees a materially low price and decides to buy ahead of the normal schedule, sometimes in larger quantities than immediately needed.

The BuyingMode label is not derived from any single official system field. Treat it as the business team's own working classification for this review, not a fixed SAP code.

The concern

Negotiators have repeatedly raised the same point in review meetings: **a low quoted price does not automatically mean a good deal.** Once you add transport cost, the cost and the physical limit of storing the material, and the risk that it loses value or expires before it is consumed, an opportunistic purchase can end up costing as much as, or more than, the regular process it was meant to beat.

The purpose of this data is to let you evaluate, order by order, whether that concern holds up - and whether it depends on what kind of material is being bought.

How to read the identifiers

Field	How to interpret it
POSampleID	One purchase order line. **PO10001-N** = purchase order 10001 received at the North Plant. The letter suffix is the receiving center: **N** = North Plant, **S** = South Plant, **P** = Port Bonded Warehouse.
Center	Receiving center as recorded in the metadata file. Also encoded in the POSampleID suffix.
DaysInStorage	How many days the batch sat in storage before it was consumed.
ShelfLifeDays	Per material. Blank for services and for materials whose group could not be identified. Note that **DaysInStorage may still be populated on those lines** by the source system - decide what to do with it and justify your choice.
StorageCostPerUnitPerDay	Cost of holding one unit for one day. Blank wherever shelf life is blank.
Reference prices	One previous price and two planned prices per material and center, with their dates. Coverage is not complete for every material-center combination.

Scope and ground rules

- The data covers purchase orders placed and delivered between **January and September 2024**.
- **purchase_metadata_SAMPLE.xlsx** is the record of what was actually bought. The other three files are reference data that hangs off it via **MaterialCode** and **Center**.
- All monetary values are in USD. Quantities, storage costs and capacities share the same normalized base unit - no conversion needed.
- Storage capacity is given as a **total** per center. Nothing tells you whether it is consumed by one order at a time or shared across concurrent batches - that is a judgement call for you to make and state.
- No cost of write-off, obsolescence or disposal is provided for material that outlives its shelf life. If your reasoning needs one, assume it and say so.

What we are looking for. Not a definitive procurement verdict. We want to see how you handle a dataset that is incomplete and partly wrong, which assumptions you make explicit, and how clearly you explain a number to someone who has to act on it.